

Forecasting the 2026 Hungarian Parliamentary Election

Polls, Fundamentals, Online Media Reach and Seat Simulation

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Why predict elections?

1.

Parties

test if strategies work

2.

Citizens

anticipate political change

3.

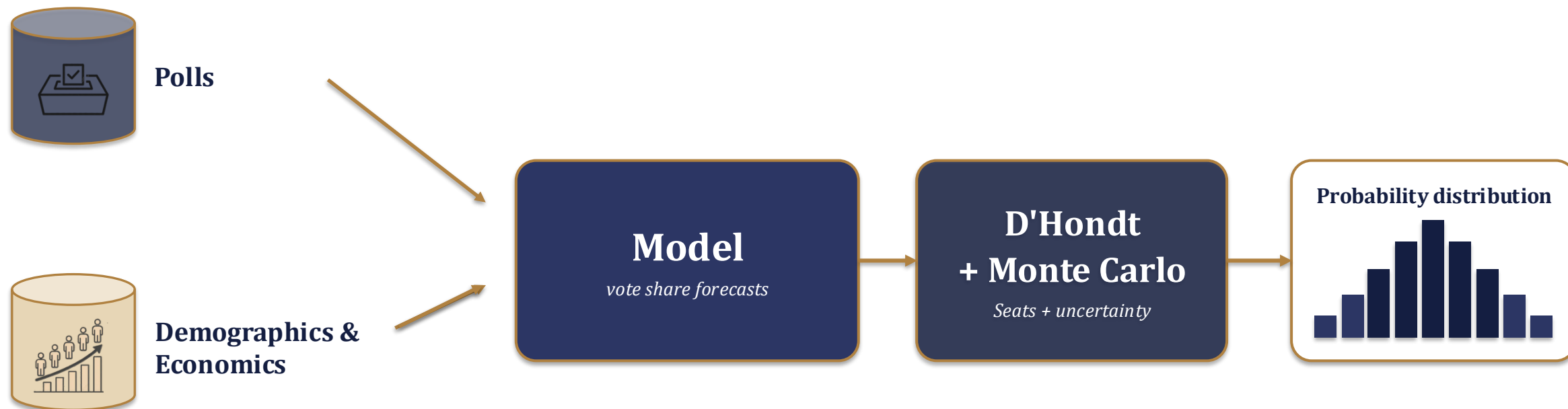
Researchers

voting behaviour

- 2026 Hungary - pre-registered results

How can we combine polls with other sources to make a good prediction about elections?

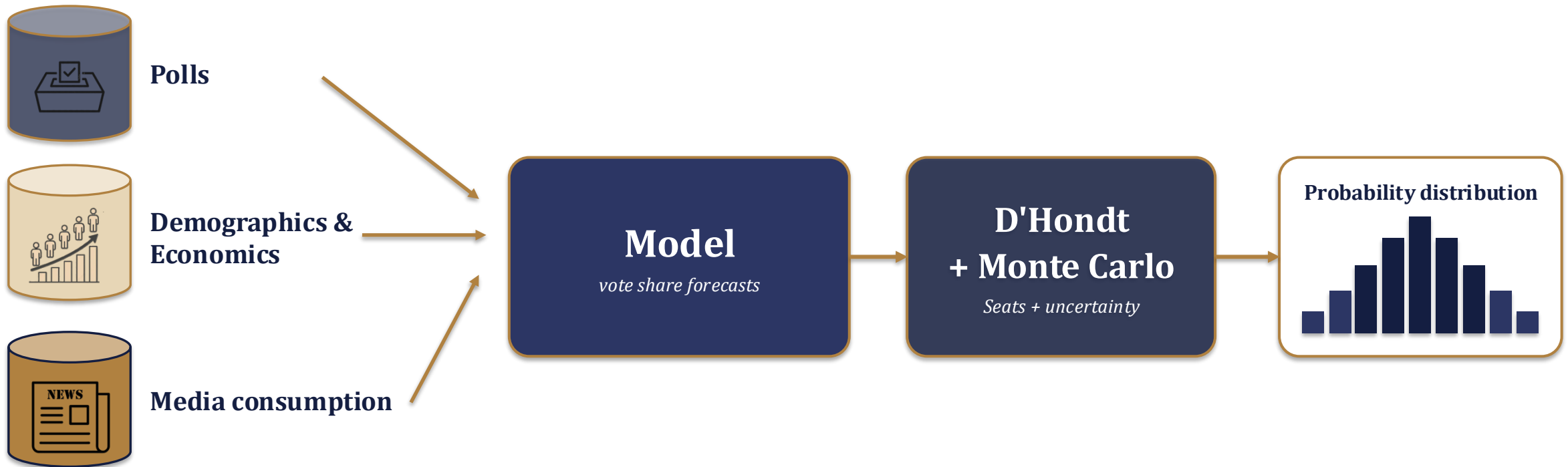
Usual forecast pipeline



**the icons are AI generated*

Based on: Linzer (2013); Stoetzer et al. (2019); Heidemanns, Gelman & Morris (2020); Erfort et al. (2026).

Usual forecast pipeline



Standard pipeline, trained on the last 2 national + 2 EP elections.

**the icons are AI generated*



Polls + Election result:

- Gábor Tóka – Vox Populi – since 2014



Economic indicators:

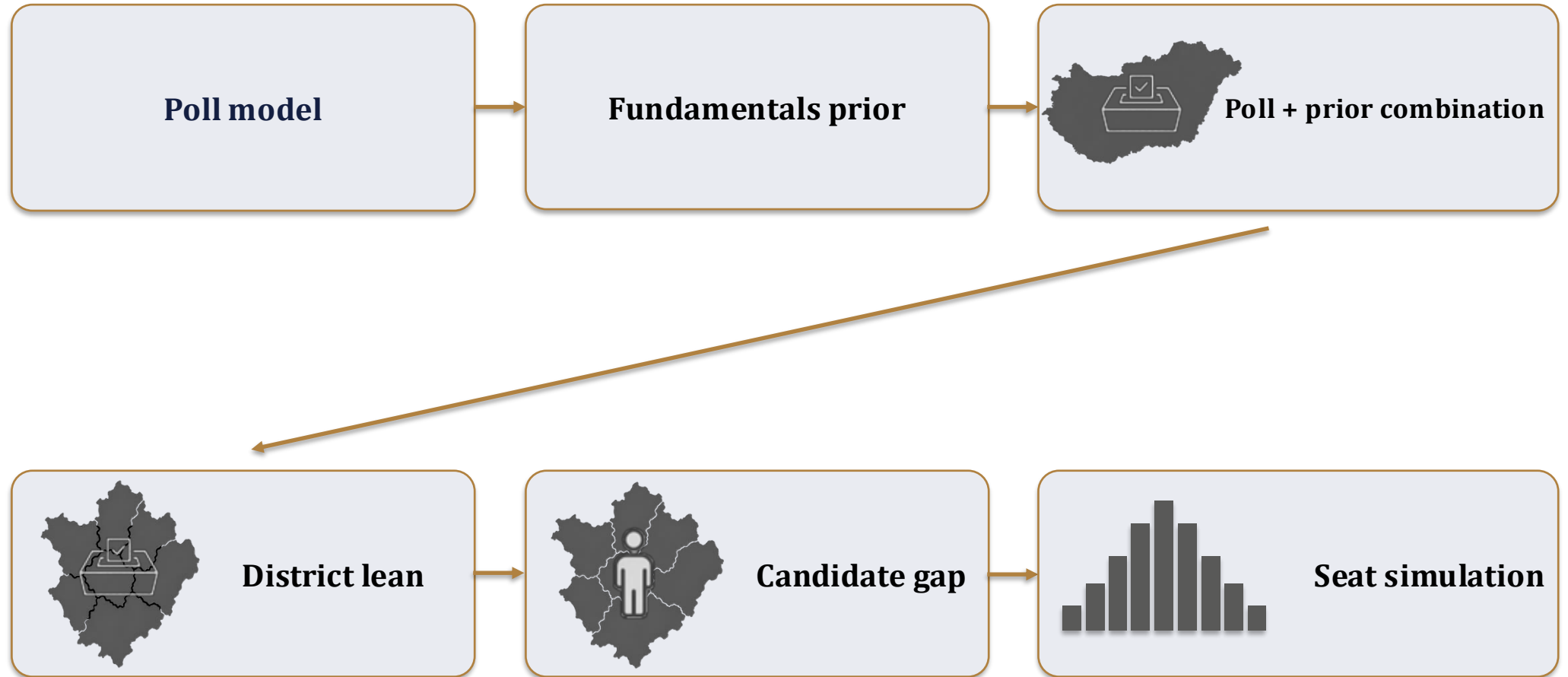
- KSH: population, unemployment, budget balance ...



Media usage:

- Gemius: telex, index, rtl, tv2 ...
- Pro-government outlets by ATLATSZO.hu
- media usage intensity

Model overview



**the icons are AI generated*

Simple model

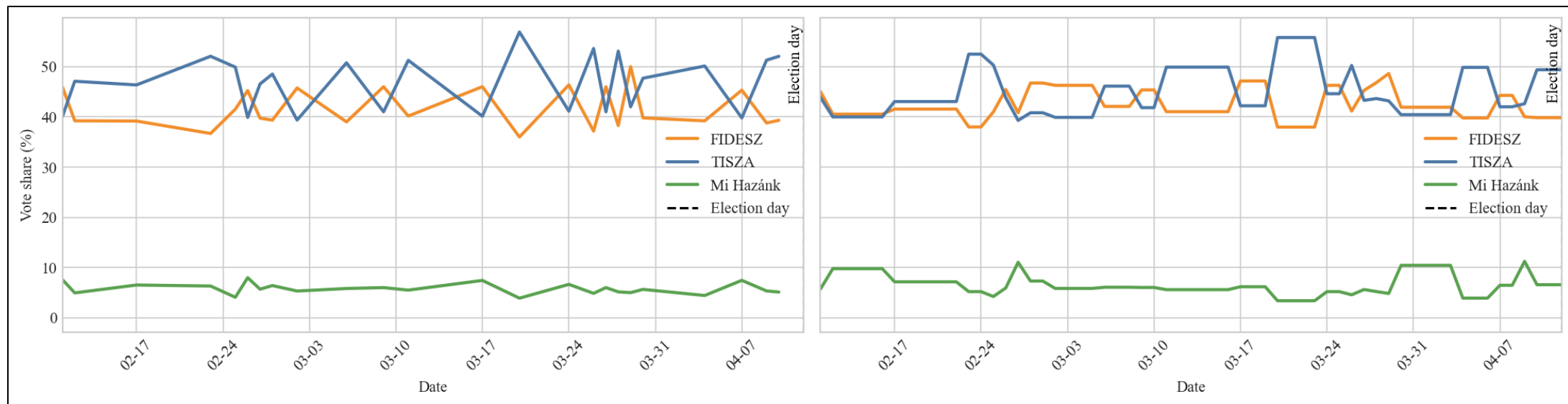
weighted average of recent polls

Advanced model

House-effects correction

Kalman filter: opinion shifts + RW

!political blocs



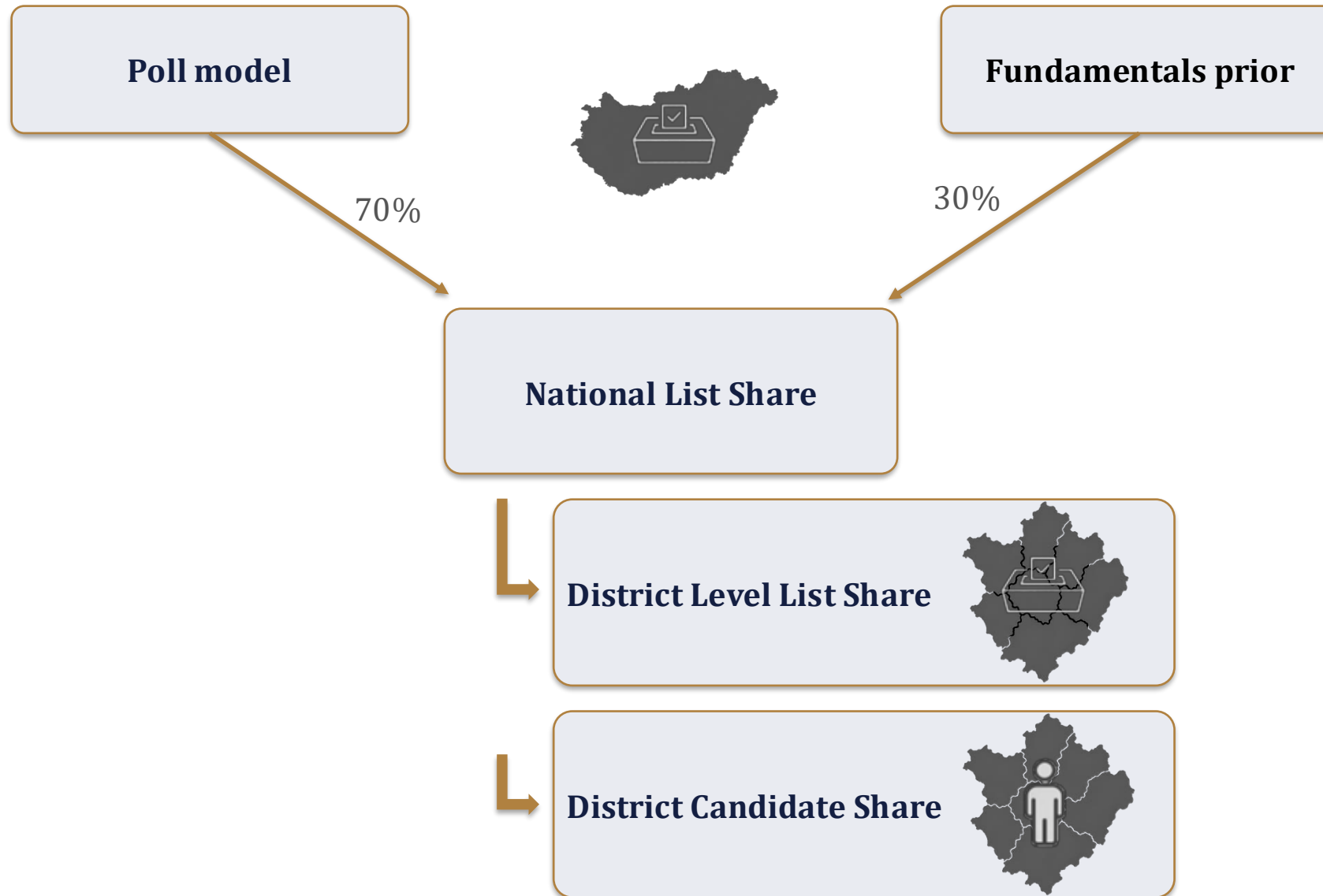
source: own editing

Which to choose?

- move slow → anchor
- population, young & old pop. share

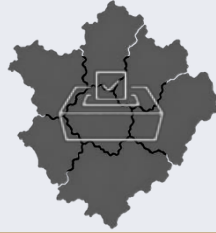
$$L(\beta_b) = \sum_{d,e} (z_{deb} - x_{de}^\top \beta_b)^2 + \alpha \sum_{j=1}^K \beta_{bj}^2$$

- district vote share → national level



■ district lean =

District Level List Share



-

National List Share



■ 1st layer:

■ demographic variables

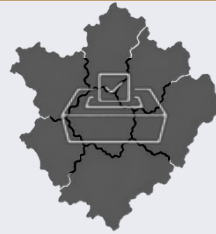
■ 2nd layer:

■ residuum = observed district lean - predicted district lean

■ media usage + past district leans (limited coverage)

■ candidate gap =

District Level List Share



-

District Candidate Share



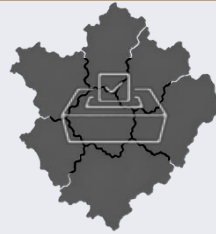
■ bloc's candidate vote $\neq$ its district list share

■ Inputs:

- past candidate gaps
- demographic variables
- district list share as anchor

■ candidate gap =

District Level List Share



-

District Candidate Share



■ bloc's candidate vote $\neq$ its district list share

■ Inputs:

- past candidate gaps
- demographic variables
- district list share as anchor

- 106 single-member districts + 93 national-list seats
- compensation votes
- D'Hondt rule → seats
- Monte Carlo (1,500 simulation) → random errors
 - district level – $N\uparrow$ – sample
 - national level & candidate – $N\downarrow$ – covariance

Pre-election forecast

Scenario	Simple poll model	70 / 30 poll + fundamentals
FIDESZ finishes first in list share	18.9%	46.6%
TISZA finishes first in list share	81.1%	53.4%
FIDESZ majority, ≥ 100 seats	18.7%	39.3%
TISZA majority, ≥ 100 seats	64.3%	29.7%
FIDESZ two thirds, ≥ 133 seats	0.1%	1.1%
TISZA two thirds, ≥ 133 seats	7.1%	1.4%
No single main bloc majority	17.0%	31.0%

Pre-registered on Zenodo before April 12.

source: own editing

Pre-election forecast vs Real results

Scenario	Fidesz	Tisza	Mi Hazánk	Other
Poll-only list prediction (%)	40.1	48.7	5.6	5.6
Real result (%)	38.6	53.2	5.6	2.6
Poll-only seat prediction (seats)	83	110	6	0
Real result (seats)	52	141	6	0

source: own editing

Direction right, magnitude wrong.

Reasons for underestimation

1.

Sea-change election

- main error: structural prior
- trained on Fidesz-era cycles
- districts
- turnout

2.

Polling quality

- underestimation
- pollster trustworthiness

3.

Garbage in, garbage out

- inputs carried errors

Generalization

- prior vs stable period
- polls (now) vs prior (past)

- **stable periods → combination**, if not **polls only**
- other sources → anchors, but 4 cycles are too few → forecast pulled away from truth
- Future directions:
 - more elections: house effects & structural prior
 - turnout → mobilization shocks

"Prediction is very difficult, especially about the future."

Niels Bohr

Thank you.

